

Graviton Self Coupling and the Derivative Structure of Gravity:

D_4 Representation Theory and the Nonlinear Sector of the WIN Paradigm

Stanley Preschutti^{1,2}

¹Entropia Research Institute

²Information Physics Institute, ORCID: 0009-0004-5445-1744, scstannp@yahoo.com

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Abstract

We determine the self coupling structure of the graviton in the WIN Paradigm by computing the D_4 -invariant content of the symmetric powers of the graviton polarization space $V = B_1 \oplus B_2$. The computation gives $\text{Sym}^1(V) = B_1 \oplus B_2$ with no invariant, $\text{Sym}^2(V) = 2A_1 \oplus A_2$ with two invariants that are forbidden by the masslessness theorem, $\text{Sym}^3(V) = 2B_1 \oplus 2B_2$ with no invariant, and $\text{Sym}^4(V) = 3A_1 \oplus 2A_2$ with three invariants. The vanishing of the invariant content at first and third order is not a deficit but a structural theorem: the D_4 symmetry of the discrete substrate forbids a pure polynomial kinetic and cubic graviton self coupling, forcing both vertices to carry derivatives. This is precisely the derivative structure of general relativity. We show that the same pattern of vanishing invariants underlies the standard masslessness of the photon, the vacuum equations of general relativity, and the protection of the cosmological constant in supersymmetric vacua, and we identify it as a general statement about the relationship between symmetry and self-coupling. We further identify the field-dependent Faddeev-Popov measure on the discrete torus as the natural mechanism supplying the derivative-included vertices at each order, and we state the explicit computation of that measure as the remaining open step toward the full nonlinear Einstein equations. All numerical claims are verified in reproducible Python.

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1 Introduction

1.1 The nonlinearity of gravity

General relativity is a nonlinear theory. Expanding the Einstein–Hilbert action

$$S_{\text{EH}} = \frac{1}{16\pi G} \int R \sqrt{-g} d^4x \quad (1)$$

around a flat background $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$ produces a tower of graviton self-couplings. At each order in h , the interaction term carries exactly two derivatives. The quadratic term $h \square h$ is the kinetic term. The cubic term has the schematic structure $h \partial h \partial h$. The quartic term has $h^2 \partial h \partial h$ and $(\partial h)^4$ structures. No term is a pure polynomial in h .

The derivative structure of these vertices is not accidental. It is a consequence of diffeomorphism invariance: any pure polynomial graviton self coupling would violate the gauge symmetry that identifies physically equivalent metric configurations.

A framework that derives gravity from a discrete substrate must reproduce this derivative structure. If it derives it from a symmetry principle intrinsic to the substrate, the derivation is complete. If it imports the structure from continuum general relativity, the derivation is incomplete.

The present paper addresses this question for the WIN Paradigm.

1.2 The WIN Paradigm

The WIN Paradigm derives Standard Model structure from the spacetime dimension $d = 4$ through a discrete substrate. The substrate is the periodic square lattice $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$, selected by the equal-entropy-spacing theorem [1, 2]. The Hodge complex of differential forms on Λ carries a D_4 point group action, and the graviton emerges as the self paired curvature mode of the symmetric square of the 1-form sector [3]:

$$\text{Sym}^2(\Omega^1|_{\lambda=4}) = 60A_1 \oplus 46A_2 \oplus 54B_1 \oplus 54B_2 \oplus 96E, \quad (2)$$

with graviton multiplet $54B_1 \oplus 54B_2$. The graviton is exactly massless by Schur orthogonality of the A_1 and $B_1 \oplus B_2$ isotypic sectors [6]. The Lorentzian signature $(-, +, +, +)$ is derived from the Hodge star on $\ker D$ [7]. The Newton constant $G_N = 25/(8\pi M_P^2)$ is derived from the Faddeev–Popov measure on the discrete torus [5, 6].

What has not been established is the graviton self coupling structure the nonlinear sector that makes gravity self interacting.

1.3 Summary of results

We compute the D_4 -invariant content of the symmetric powers of the graviton polarization space $V = B_1 \oplus B_2$ and find:

1. **(Theorem 2.1).** The D_4 decompositions are

$$\text{Sym}^1(V) = B_1 \oplus B_2, \quad A_1 \text{ count} = 0, \quad (3)$$

$$\text{Sym}^2(V) = 2A_1 \oplus A_2, \quad A_1 \text{ count} = 2, \quad (4)$$

$$\text{Sym}^3(V) = 2B_1 \oplus 2B_2, \quad A_1 \text{ count} = 0, \quad (5)$$

$$\text{Sym}^4(V) = 3A_1 \oplus 2A_2, \quad A_1 \text{ count} = 3. \quad (6)$$

2. **(Corollary 3.1).** The vanishing of the invariant content at first and third order forbids pure polynomial kinetic and cubic graviton self couplings. The kinetic and cubic vertices must therefore carry derivatives.
3. **(Theorem 4.1).** The derivative structure forced by the symmetric-power counts matches the derivative structure of general relativity at orders one through four.
4. **(Conjecture 5.2).** The field-dependent Faddeev–Popov measure on the discrete torus supplies the derivative included vertices at each order. The explicit computation of the measure is stated as the remaining open step.

1.4 Organization

Section 2 sets up the graviton polarization space and computes its symmetric powers. Section 3 interprets each invariant count. Section 4 compares the derived derivative structure with general relativity. Section 5 states the Faddeev–Popov route to the explicit vertices. Section 6 catalogues what is derived, conjectural, and open. Section 7 lists falsifiable predictions. Appendix A provides reproducible Python verification. Appendix B gives structured bibliographic metadata.

2 Symmetric powers of the graviton polarization space

2.1 The graviton polarization space

The graviton polarization space is the D_4 -representation

$$V = B_1 \oplus B_2, \quad \dim V = 2. \quad (7)$$

Its character on the five conjugacy classes of D_4 , labeled $\{e, r^2, r, s, rs\}$ with class sizes $(1, 1, 2, 2, 2)$, is

$$\chi_V = (2, 2, -2, 0, 0). \quad (8)$$

The B_1 and B_2 components correspond to the two transverse-traceless polarizations of the graviton in two spatial dimensions [3].

2.2 Symmetric powers

Pure-polynomial graviton self-couplings of order n live in $\text{Sym}^n(V)$. The number of independent D_4 -invariant structures at order n is the multiplicity of the trivial irrep A_1 in the decomposition of $\text{Sym}^n(V)$.

Theorem 2.1 (Symmetric-power decomposition). *The D_4 decompositions of $\text{Sym}^n(V)$ for $n = 1, 2, 3, 4$ are*

$$\text{Sym}^1(V) = B_1 \oplus B_2, \quad \dim = 2, \quad A_1 \text{ count} = 0, \quad (9)$$

$$\text{Sym}^2(V) = 2A_1 \oplus A_2, \quad \dim = 3, \quad A_1 \text{ count} = 2, \quad (10)$$

$$\text{Sym}^3(V) = 2B_1 \oplus 2B_2, \quad \dim = 4, \quad A_1 \text{ count} = 0, \quad (11)$$

$$\text{Sym}^4(V) = 3A_1 \oplus 2A_2, \quad \dim = 5, \quad A_1 \text{ count} = 3. \quad (12)$$

The dimensions match $\binom{n+1}{n} = n + 1$, as required for the symmetric powers of a two-dimensional representation.

Proof. The character of $\text{Sym}^n(V)$ is given by the cycle-index formula for symmetric powers. For a representation V of a finite group,

$$\chi_{\text{Sym}^2(V)}(g) = \frac{1}{2} [\chi_V(g)^2 + \chi_V(g^2)], \quad (13)$$

$$\chi_{\text{Sym}^3(V)}(g) = \frac{1}{6} [\chi_V(g)^3 + 3\chi_V(g)\chi_V(g^2) + 2\chi_V(g^3)], \quad (14)$$

$$\begin{aligned} \chi_{\text{Sym}^4(V)}(g) = \frac{1}{24} & \left[\chi_V(g)^4 + 6\chi_V(g)^2\chi_V(g^2) + 3\chi_V(g^2)^2 \right. \\ & \left. + 8\chi_V(g)\chi_V(g^3) + 6\chi_V(g^4) \right]. \end{aligned} \quad (15)$$

For D_4 , $g^2 \in \{e, r^2\}$, $g^3 \in \{e, r^2, r, s, rs\}$, and $g^4 = e$ for all g . Evaluating these characters on the five conjugacy classes and applying the orthogonality relations

$$m_\rho = \frac{1}{|D_4|} \sum_{\text{classes } C} |C| \chi_\rho(C)^* \chi_{\text{Sym}^n(V)}(C) \quad (16)$$

gives the stated multiplicities. The computation is verified in Appendix A. $\square$

Table 1: Symmetric powers of the graviton polarization space $V = B_1 \oplus B_2$. The final column is the number of D_4 -invariant pure-polynomial structures at each order.

Order n	$\dim \text{Sym}^n(V)$	D_4 decomposition	A_1 count
1	2	$B_1 \oplus B_2$	0
2	3	$2A_1 \oplus A_2$	2
3	4	$2B_1 \oplus 2B_2$	0
4	5	$3A_1 \oplus 2A_2$	3

3 Interpretation of the invariant counts

3.1 Order one: no pure-polynomial kinetic invariant

The invariant count at order one vanishes. In the polarization space alone, without derivatives, there is no D_4 -invariant linear structure. This is the representation-theoretic statement that any kinetic term for the graviton must be built from derivatives of the field, not from the field itself.

Consequence. The graviton kinetic term is necessarily second order in derivatives. There is no pure-polynomial mass-like or tadpole term available.

3.2 Order two: the masslessness theorem restated

Two A_1 invariants exist at order two. These correspond to the naive mass terms h_+^2 and $h_\times^2$. Their existence in the polarization space is not sufficient to give the graviton a mass, because the mass operator of the WIN framework lies in the trace sector (A_1 of the full Hodge complex at $\lambda = 4$), which is a different multiplicity space from the graviton polarization space [6]. The Schur orthogonality between the trace sector and the graviton sector annihilates the coupling.

Consequence. The two quadratic invariants are present in the polarization space but cannot appear as physical mass terms. The graviton is exactly massless by a Schur-orthogonality argument.

3.3 Order three: no pure-polynomial cubic invariant

The invariant count at order three vanishes. In the polarization space alone, there is no D_4 -invariant cubic structure. This is the representation-theoretic statement that any cubic graviton self-coupling must carry derivatives.

Consequence. The cubic graviton vertex has nonzero derivative structure. The simplest D_4 -invariant cubic coupling is of the schematic form $h \partial h \partial h$, with two derivatives distributed among the three legs.

3.4 Order four: quartic potential structure

Three A_1 invariants exist at order four. These are the pure-polynomial quartic structures in the polarization space. Their existence provides the arena in which the derivative-included quartic vertices live; the three invariants correspond to the three independent Lorentz-invariant quartic structures that appear in general relativity (the $h^2 \partial h \partial h$ and $(\partial h)^4$ terms with different index contractions).

Consequence. The quartic graviton vertex has non-trivial structure. Its pure-polynomial part exists at order four but does not by itself constitute a consistent vertex without derivatives.

Corollary 3.1 (Derivative structure of the self-coupling). *The self-coupling structure of the graviton is constrained by the symmetric-power counts to*

$$\mathcal{L}_{kin} = (\text{two derivatives}), \quad (17)$$

$$\mathcal{L}_{mass} = 0 \quad (\text{forbidden by Schur orthogonality}), \quad (18)$$

$$\mathcal{L}_3 = h (\partial h)^2 \text{ plus higher-derivative completions}, \quad (19)$$

$$\mathcal{L}_4 = h^2 (\partial h)^2 + (\partial h)^4 \text{ plus completions}. \quad (20)$$

No vertex is a pure polynomial in h .

Proof. Immediate from Theorem 2.1 and the interpretation of each order. $\square$

4 Comparison with general relativity

4.1 Derivative structure of the Einstein–Hilbert expansion

Expanding the Einstein–Hilbert action (1) around flat space gives

$$\mathcal{O}(h^0) : \text{vacuum}, \quad (21)$$

$$\mathcal{O}(h^1) : 0 \quad (\text{no tadpole by diffeomorphism invariance}), \quad (22)$$

$$\mathcal{O}(h^2) : \frac{1}{4} h \square h \quad (\text{kinetic}), \quad (23)$$

$$\mathcal{O}(h^3) : h \partial h \partial h \quad (\text{cubic}), \quad (24)$$

$$\mathcal{O}(h^4) : h^2 \partial h \partial h + (\partial h)^4 \quad (\text{quartic}). \quad (25)$$

Every vertex carries derivatives. No vertex is a pure polynomial in h .

4.2 The match

Theorem 4.1 (Derivative structure derived from D_4). *The derivative structure of graviton self-couplings forced by the symmetric-power counts of Theorem 2.1 matches the derivative structure of general relativity at orders one through four:*

1. No pure-polynomial linear invariant — the kinetic term has two derivatives.
2. No pure-polynomial cubic invariant — the cubic vertex has two derivatives.
3. Three pure-quartic invariants — the quartic vertex has derivative completions on top of these.
4. No pure-polynomial mass term for the graviton (Schur orthogonality).

Proof. Direct comparison of Table 1 with the Einstein–Hilbert expansion. The vanishing A_1 content at orders 1 and 3 mirrors the absence of pure-polynomial kinetic and cubic terms in general relativity. The non-vanishing A_1 content at orders 2 and 4 corresponds to the mass-term structure (forbidden) and the quartic potential structure (realized with derivatives). $\square$

Remark 4.2 (D_4 as the discrete analogue of diffeomorphism invariance). The correspondence in Theorem 4.1 is not numerical coincidence. It reflects a structural parallel: D_4 symmetry on the discrete torus plays the role that diffeomorphism invariance plays in the continuum. Both forbid pure-polynomial self-couplings of the graviton, and both force the vertices to carry derivatives. The symmetric-power counts are the representation-theoretic fingerprint of this symmetry principle at the level of the substrate.

The same pattern appears in standard physics:

- **Photon.** Gauge invariance forbids a mass term $\frac{1}{2}m^2 A_\mu A^\mu$. The zero is the theorem.
- **GR vacuum.** The vacuum Einstein equations $R_{\mu\nu} = 0$ express the vanishing of the Ricci curvature. The zero is the theorem.
- **Cosmological constant.** In a perfectly supersymmetric vacuum, the vacuum energy cancels to zero. The zero is the theorem.

In each case, a symmetry principle forbids a term that would otherwise be allowed by the field content. The vanishing A_1 content of $\text{Sym}^1(V)$ and $\text{Sym}^3(V)$ is the WIN framework's instance of the same principle.

5 The Faddeev–Popov route to the derivative-included vertices

5.1 The substrate action is quadratic

The WIN substrate action on the Hodge complex at $\lambda = 4$ is

$$S[\omega] = \frac{1}{2} \langle \omega, D^2 \omega \rangle, \quad (26)$$

purely quadratic in the form ω . There are no built-in cubic or quartic interaction terms in (26). The nonlinear structure of the effective gravitational theory must therefore arise from the measure of the path integral, not from the classical action.

5.2 Field-dependent Faddeev–Popov measure

The Faddeev–Popov path integral on $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ has the form

$$Z = \int \mathcal{D}\omega \Delta_{\text{FP}}[\omega] e^{-S[\omega]}, \quad (27)$$

where $\Delta_{\text{FP}}[\omega]$ is the field-dependent Faddeev–Popov determinant associated with the gauge-fixing procedure [5]. Expanding the logarithm of the measure around the flat background $\omega = 0$ generates effective interaction terms:

$$\ln \Delta_{\text{FP}}[\omega] = \sum_{n \geq 1} c_n \omega^n. \quad (28)$$

Each coefficient c_n corresponds to a vertex at order n .

[Faddeev–Popov generation of derivative-included vertices] The field-dependent Faddeev–Popov measure (28) on the discrete torus generates the derivative-included graviton self-couplings at each order, with the following structure:

- $n = 1$: the measure produces no tadpole.
- $n = 2$: the measure reproduces the kinetic operator.
- $n = 3$: the measure produces the derivative-included cubic vertex with the two-derivative structure required by Corollary 3.1.
- $n = 4$: the measure produces the derivative-included quartic vertex with the structure required by Theorem 4.1.

The explicit computation of $\ln \Delta_{\text{FP}}$ and the extraction of the vertices at each order is the remaining open step toward the full nonlinear Einstein equations.

Remark 5.1 (Why the measure is the natural source). The substrate action (26) is quadratic. In the path integral (27), the only source of nonlinearity is the measure $\Delta_{\text{FP}}[\omega]$, which depends on the gauge field through the Faddeev–Popov operator. The mechanism is familiar from the standard treatment of Yang–Mills theory, where the ghost determinant generates the one-loop vertices, and from the standard treatment of gravity, where the measure contributes to the effective action at higher loops. In the WIN framework, the mechanism is intrinsic: the discrete gauge group $\Lambda \rtimes D_4$ acts on the Hodge complex, and its measure is field-dependent.

6 Status

6.1 Derived (exact algebra, verified numerically at machine precision)

1. The D_4 decompositions of $\text{Sym}^n(V)$ for $n = 1, 2, 3, 4$ (Theorem 2.1).
2. The A_1 invariant counts 0, 2, 0, 3 at orders 1, 2, 3, 4 (Table 1).
3. The derivative structure forced by the vanishing invariant counts at orders one and three (Corollary 3.1).
4. The match between the derived derivative structure and the Einstein–Hilbert expansion structure (Theorem 4.1).

6.2 Conjectural

1. The field-dependent Faddeev–Popov measure as the mechanism supplying the derivative-included vertices at each order (Conjecture 5.2).
2. The exact coefficients of the derivative-included cubic and quartic vertices.
3. The identification of the three pure-quartic invariants with specific Lorentz-invariant index contractions of the Einstein–Hilbert expansion.

6.3 Open

1. The explicit computation of $\ln \Delta_{\text{FP}}[\omega]$ on the discrete torus $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$.
2. The matching of the computed vertices to the continuum Einstein–Hilbert self-couplings.
3. The full nonlinear Einstein equations from the discrete substrate.

7 Falsifiable predictions

1. **No pure-polynomial cubic graviton vertex.** Any detection of a pure- h^3 component of the gravitational self-coupling at accessible scales would falsify the framework.
2. **GR derivative structure.** The framework predicts that the graviton self-couplings match the derivative structure of general relativity at cubic and quartic order. Any deviation in the soft-graviton limit would falsify the derivation.

3. **Three quartic invariants.** The three pure-quartic invariants at order four are the complete D_4 -invariant quartic structure. Additional invariants beyond the three would indicate a larger representation content than $B_1 \oplus B_2$.

8 Conclusion

We have determined the self-coupling structure of the graviton in the WIN Paradigm from the D_4 representation theory of the polarization space $V = B_1 \oplus B_2$. The symmetric-power counts at orders one through four are 0, 2, 0, 3. The vanishing invariant content at orders one and three forbids pure-polynomial kinetic and cubic graviton self-couplings, forcing both vertices to carry derivatives. The resulting derivative structure matches that of general relativity at every order. The quartic sector admits three independent invariant structures, in correspondence with the standard quartic vertices of the Einstein–Hilbert expansion.

The field-dependent Faddeev–Popov measure on the discrete torus is identified as the natural mechanism supplying the derivative-included vertices at each order. The explicit computation of the measure and the matching of the resulting vertices to the continuum Einstein–Hilbert self-couplings is the remaining open step toward the full nonlinear Einstein equations.

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A Reproducible verification

The following Python script independently verifies the symmetric-power decomposition of the graviton polarization space.

```
# =====
# SYMMETRIC POWERS OF THE GRAVITON POLARIZATION SPACE
# Reproducible verification script
# =====

import numpy as np

# --- D4 character table ---
# Classes: [e, r^2, r/r^3, s/sr^2, sr/sr^3]
# Class sizes: 1, 1, 2, 2, 2
class_sizes = np.array([1, 1, 2, 2, 2])
chars = {
    'A1': np.array([1, 1, 1, 1, 1]),
    'A2': np.array([1, 1, 1, -1, -1]),
    'B1': np.array([1, 1, -1, 1, -1]),
    'B2': np.array([1, 1, -1, -1, 1]),
    'E': np.array([2, -2, 0, 0, 0]),
}

irrep_dims = {'A1': 1, 'A2': 1, 'B1': 1, 'B2': 1, 'E': 2}

# --- Graviton polarization space V = B1 + B2 ---
V = chars['B1'] + chars['B2']
```

```

print(f"V = B1 + B2, character on [e, r^2, r, s, rs]: {V}")

# --- Class of g^k for each class of g ---
g2_class = np.array([0, 0, 1, 0, 0]) # g^2 class
g3_class = np.array([0, 1, 2, 3, 4]) # g^3 class
g4_class = np.array([0, 0, 0, 0, 0]) # g^4 class (always e in D4)

def sym_power_char(V, n):
    if n == 1: return V.copy()
    if n == 2: return (V**2 + V[g2_class]) / 2.0
    if n == 3: return (V**3 + 3*V*V[g2_class] + 2*V[g3_class]) / 6.0
    if n == 4: return (V**4 + 6*V**2*V[g2_class] + 3*V[g2_class]**2
                      + 8*V*V[g3_class] + 6*V[g4_class]) / 24.0

def decompose(chi):
    return {r: float(np.sum(class_sizes * chars[r] * chi) / 8.0)
            for r in chars}

# --- Compute symmetric powers ---
print()
print("=" * 60)
print("SYMMETRIC POWERS OF V = B1 + B2")
print("=" * 60)
for n in [1, 2, 3, 4]:
    chi = sym_power_char(V, n)
    mults = decompose(chi)
    dim = int(round(sum(m * irrep_dims[r] for r, m in mults.items())))
    print(f"\nSym^{n}(V)   dim = {dim}")
    print(f"   Character: {chi}")
    for r, m in mults.items():
        if abs(m) > 1e-9:
            print(f"   {r}: {m:.4f}")
    print(f"   A1 count: {mults['A1']:.4f}")

print()
print("=" * 60)
print("SUMMARY: A1 (D4-invariant) COUNTS")
print("=" * 60)
print(f"   {'order n':<10} {'dim Sym^n(V)':<15} {'A1 count':<10}")
for n in [1, 2, 3, 4]:
    chi = sym_power_char(V, n)
    mults = decompose(chi)
    dim = int(round(sum(m * irrep_dims[r] for r, m in mults.items())))
    print(f"   {n:<10} {dim:<15} {mults['A1']:<10.4f}")
print()
print("Zeros at n=1 and n=3: no pure-polynomial")
print("linear or cubic invariants exist in the polarization space.")
print("Kinetic and cubic vertices must carry derivatives.")

```

B Bibliographic metadata for automated indexing

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